Robust Multi-Modal Perception and State Estimation for Robotic Guide Dogs: An Adaptive Heterogeneous Physical Feature Fusion and Semantic-Adaptive Particle Filter Approach

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Abstract

Robotic guide dogs must perceive terrain semantics and maintain a reliable 6-DoF pose estimate in unstructured environments where cameras and depth sensors can degrade abruptly. This paper presents a practical perception-and-estimation framework that couples an Adaptive Heterogeneous Physical Feature Fusion (AHPFF) front end with a Semantic-Adaptive Particle Filter (SAPF) back end. AHPFF extracts hand-crafted visual descriptors (Canny edge density, Laplacian variance, intensity and gradient statistics, entropy, and color statistics) and depth descriptors (depth statistics and Sobel gradients). An adaptive silhouette-weighted fusion is used in the clustering ablation, while a lightweight supervised semantic readout converts the AHPFF descriptor bank into a discrete terrain label $C_t \in \{\text{Flat}, \text{Stairs}, \text{Obstacle}, \text{Tactile}\}$. The SAPF tracks the state $[p_x, p_y, p_z, \phi, \theta, \psi]^T$ with a class-dependent process covariance $\mathbf{Q}(C_t)$ and systematic resampling. On a corrected NYU Depth V2 pilot subset (15 samples per class, limited by the small number of annotated stair frames), AHPFF reaches 71.1% clustering accuracy and 36.7% NMI under multi-modal degradation, compared with 66.7%/28.1% for fixed concatenation and 51.1%/10.0% for vision alone. In a ROS/Gazebo closed-loop simulation with a forward RGB-D Go1 and a dynamic pedestrian, the semantic readout reaches 90.1% accuracy, and SAPF reduces the Z-axis RMSE to 0.024 m and the overall position RMSE to 0.030 m, compared with 0.150 m and 0.152 m for a standard particle filter. Under software-injected overexposure, motion blur, and 50% depth dropout, the semantic accuracy drops from 90.8% to 71.5%, yet SAPF keeps the Z-axis RMSE at 0.022 m while the standard particle filter degrades from 0.154 m to 0.190 m. The Go1 motion in the Gazebo study is kinematic; no real-world deployment is claimed.

Keywords: Multi-modal fusion, Hand-crafted physical descriptors, Semantic-adaptive particle filter, Robotic guide dog, 6-DoF state estimation

1. Introduction

Robotic guide dogs are expected to accompany visually impaired users through sidewalks, tactile paving, obstacles, and staircases. Unlike vehicles operating on well-structured roads, a quadrupedal guide dog must interpret fine-grained terrain semantics while maintaining a reliable 6-DoF pose estimate in the presence of locomotion-induced vibration, foot slip, and sensor degradation. A single sensing modality is fragile in this setting: vision is affected by overexposure, motion blur, and illumination changes, whereas depth or LiDAR measurements can become sparse or corrupted by multipath reflections and dropouts.

Multi-modal fusion is therefore necessary, but a fixed fusion rule is not sufficient. When one modality is severely degraded, naive concatenation or fixed weighting can propagate corrupted features into the downstream estimator, a failure often referred to as negative transfer. In addition, the semantic label produced by the perception front end is rarely connected to the process noise of the state estimator: a conventional particle filter uses the same motion covariance on flat ground and on stairs, which can cause delayed elevation response or z-axis drift.

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This paper addresses these issues with two coupled components. The first is Adaptive Heterogeneous Physical Feature Fusion (AHPFF), a hand-crafted descriptor pipeline that combines visual texture and depth geometry without resource-intensive deep networks. AHPFF uses silhouette confidence to weight the two views in the clustering ablation, and a lightweight supervised readout to provide a terrain label C_t in the online loop. The second is a Semantic-Adaptive Particle Filter (SAPF) that tracks the 6-DoF state and modulates its process covariance according to C_t: stair labels expand vertical and pitch uncertainty, while obstacle labels expand horizontal and yaw uncertainty. The filter uses systematic resampling and is evaluated against wheel odometry (WO) and a standard particle filter (SPF) with a fixed covariance.

The main contributions are:

- We formulate AHPFF with hand-crafted visual and depth descriptors, a silhouette-weighted PCA fusion for the clustering ablation, and a lightweight semantic readout for online terrain labeling. The design avoids deep feature extractors and remains interpretable.
- We implement a 6-DoF SAPF with a class-dependent covariance $\mathbf{Q}(C_t)$ and systematic resampling. The filter is integrated with a forward RGB-D Go1 in ROS/Gazebo and evaluated against WO and SPF baselines.
- We provide a corrected NYU Depth V2 pilot study, a Gazebo closed-loop study with a dynamic pedestrian and scripted human handler, and a controlled sensor-degradation study. The results quantify both the benefit and the limitations of the current implementation: semantic accuracy drops under severe degradation, but SAPF maintains a stable vertical estimate.

2. Related Work

2.1. Sensor-Driven Assistive Navigation and Guide Dog Robots

Sensor-driven assistive technologies have been developed around smartphones, electronic white canes, and wearable devices [1, 2, 3]. The Virtual Guide Dog (VGD) uses onboard sensors and the NTCIP protocol to communicate with traffic controllers at intersections [4]. With the progress of legged robotics, quadrupedal platforms have been explored as physical guide dogs. The iDog system uses AI and edge computing for real-time audio alerts [5], and Hwang et al. developed a guide dog robot with a rigid harness and a semantics-aware local planner [6].

Physical human-robot interaction and locomotion comfort are also critical. Fan et al. proposed a force-compliance MPC framework for interactive navigation and obstacle avoidance [7], while Yu et al. emphasized quiet and stable locomotion for acoustic situational awareness [8]. These systems demonstrate the importance of robust perception and state estimation, but they typically use deterministic or loosely coupled fusion and do not explicitly connect terrain semantics to the process noise of a probabilistic filter.

2.2. Multi-View Clustering and Feature Fusion

Multi-view clustering (MVC) provides a natural formulation for combining heterogeneous sensor views [9, 10, 11]. Graph-regularized nonnegative matrix factorization embeds local geometric structure and orthogonal constraints [12]; spectral analysis with local refinement captures fine-grained visual topology [13]; and angular metric learning can produce tighter class boundaries [14]. Continual and incomplete-view methods update consensus coefficients or handle missing samples [15, 16], and auto-weighted binary MVC learns view and sample weights jointly [17].

Most of these methods are designed for offline or soft-real-time clustering. In this work, AHPFF uses a compact set of hand-crafted descriptors, applies silhouette confidence to weight the views in the clustering ablation, and relies on a lightweight supervised readout for the online label. The online readout is intentionally separated from the clustering ablation: with a small terrain calibration set, a supervised classifier is more reliable than assigning cluster indices to classes.

2.3. Semantic-Aware State Estimation

Particle filters are widely used for non-linear, multi-modal state estimation [18]. In legged and assistive robots, process and observation noise are often tuned for a single terrain type, which is inadequate when the robot transitions between flat ground, tactile paving, obstacles, and stairs. Semantic-aware estimation addresses this by using a terrain label to modulate the motion or observation model. Our SAPF follows this idea in a deliberately simple form: a discrete terrain label changes the diagonal process covariance, and the filter uses systematic resampling with an effective-sample-size trigger. The observation model in the Gazebo study is a simulated noisy absolute pose, which serves as a stand-in for an exteroceptive localization module.

3. Methodology

The framework consists of an AHPFF front end and a 6-DoF SAPF back end. The front end produces a terrain label C_t; the back end uses C_t to select the process covariance.

3.1. Adaptive Heterogeneous Physical Feature Fusion (AHPFF)

Let $\mathbf{X}_t^{(1)}$ and $\mathbf{X}_t^{(2)}$ denote the visual and depth observations. AHPFF extracts interpretable descriptors rather than learned embeddings. The visual descriptor includes Canny edge density, Laplacian variance, mean and standard deviation of intensity, mean and standard deviation of gradient magnitude, intensity entropy, and color statistics (mean RGB, saturation, and yellow/red fractions in the online RGB-D version). The depth descriptor includes mean, standard deviation, minimum, maximum, percentiles, depth span, mean absolute Sobel responses, and mean gradient magnitude.

The two descriptor vectors are standardized and projected by PCA to a common latent dimension. The fused representation is

$$\mathbf{Z}_t^f = \alpha \mathbf{Z}_t^{(1)} + (1-\alpha)\mathbf{Z}_t^{(2)}, \tag{1}$$

where $\alpha = s_1/(s_1 + s_2)$ and s_1, s_2 are the silhouette scores of the two views with respect to a consensus partition. This fusion is used in the clustering ablation. For the online loop, the AHPFF descriptor bank is fed to a lightweight ExtraTrees classifier trained on labeled calibration samples; the classifier returns $C_t \in \{\text{Flat}, \text{Stairs}, \text{Obstacle}, \text{Tactile}\}$. The online semantic readout is therefore supervised, while silhouette weighting is used only in the offline ablation. This distinction is stated explicitly to avoid presenting the online classifier as an unsupervised clustering step.

3.2. Semantic-Adaptive Particle Filter (SAPF)

The state is $\mathbf{x}_t = [p_x, p_y, p_z, \phi, \theta, \psi]^T$, where ϕ, θ, ψ are roll, pitch, and yaw. The filter maintains N particles $\{\mathbf{x}_t^{[i]}, w_t^{[i]}\}_{i=1}^N$.

3.2.1. Motion Model and Semantic Covariance

The state transition is

$$\mathbf{x}_t^{[i]} = \mathbf{x}_{t-1}^{[i]} + \mathbf{u}_t + \boldsymbol{\epsilon}_t(C_t), \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}(C_t)), \tag{2}$$

where $\mathbf{u}_t$ is the odometry increment. The process covariance is

$$\mathbf{Q}(C_t) = \mathbf{Q}_{base} + \operatorname{diag}\left(\boldsymbol{\beta}_{C_t} \odot \boldsymbol{\sigma}^2_{max}\right), \tag{3}$$

where $\mathbf{Q}_{base}$ is the nominal covariance and $\boldsymbol{\beta}_{C_t}$ is a configured activation vector. The boosts are calibrated from the stair geometry and sensor noise rather than learned end-to-end. For stairs, the vertical and pitch components are increased; for obstacles, the horizontal and yaw components are increased; tactile paving receives a small increase. A standard particle filter (SPF) uses $\mathbf{Q}_{base}$ for every label.

3.2.2. Observation Model and Resampling

The observation is a 6-DoF pose measurement

$$\mathbf{z}_t = \mathbf{x}_t + \mathbf{n}_t, \quad \mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \tag{4}$$

which is used to update the importance weights:

$$w_t^{[i]} = \eta \, p(\mathbf{z}_t \mid \mathbf{x}_t^{[i]}) \, w_{t-1}^{[i]}. \tag{5}$$

The likelihood is Gaussian in position and orientation, with angles wrapped to $[-\pi, \pi]$. In the Gazebo experiments, $\mathbf{z}_t$ is a simulated noisy absolute pose; it is a stand-in for an exteroceptive localization module rather than a latent-feature likelihood.

The effective sample size is

$$\mathrm{ESS}_t = \frac{1}{\sum_{i=1}^{N} (w_t^{[i]})^2}. \tag{6}$$

When $\mathrm{ESS}_t < N/2$, systematic resampling is applied. With $u_0 \sim \mathcal{U}[0, 1/N)$ and $u_i = u_0 + i/N$, particle i is selected according to the cumulative weight interval containing u_i. The state estimate uses the weighted mean for positions and the circular mean for angles.

3.3. ROS/Gazebo Integration

The Go1 is equipped with a forward RGB-D camera. The perception node extracts the AHPFF descriptors from the color and depth images, vertically flips the raw camera image in software, and publishes C_t at approximately 10 Hz. The SAPF node runs at 50 Hz with 500 particles, reads the robot pose from Gazebo model states, simulates noisy odometry and a noisy absolute pose observation, and publishes the filtered pose. The street world contains a concrete sidewalk, curb and road, brick facade, tactile paving, a ten-step staircase (0.15 m rise, 0.30 m run), a bollard, street furniture, a static pedestrian, a walking pedestrian actor, and a scripted human handler. The Go1 model state is commanded along a scripted trajectory; its legs are animated, but the motion is not generated by a physics-based locomotion controller. The human handler is a gravity-disabled, non-colliding model that is teleported with the robot. These choices are stated explicitly because they bound the interpretation of the Gazebo results.

4. Experiments

4.1. AHPFF Ablation on a Corrected NYU Depth V2 Pilot Subset

The original implementation used NYU label ids 1/42/4, which correspond to book/shelves/ceiling in the 1-based NYU annotation and are not terrain classes. We corrected the mapping to floor=11, stairs=215, and chair=5 (obstacle proxy). NYU Depth V2 contains only 17 frames with stair annotations; with a minimum mask area of 500 pixels, 15 frames remain. The corrected balanced subset therefore contains 15 samples per class (45 samples in total). This is a pilot subset, not a 200-samples-per-class benchmark.

We extract object-masked Canny, Laplacian, and Sobel descriptors, inject visual degradation in 20–40% of the samples and depth/LiDAR degradation in 60–80% of the samples, and compare the two single views, fixed concatenation, and AHPFF. Table 1 and Fig. 1 report the results. AHPFF achieves the best degraded accuracy and NMI. On clean data, however, AHPFF is not the best method (60.0% ACC versus 62.2% for depth alone); the advantage appears under the injected degradation, where the adaptive view weighting suppresses the corrupted stream.

4.2. Go1 Closed-Loop Study in a Street Scene

We evaluate the full AHPFF–SAPF loop on a Go1 with a forward RGB-D camera in the street world. The perception node is calibrated on 351 samples collected from the same scene; an ExtraTrees classifier achieves 98.29% five-fold cross-validation accuracy. The online semantic accuracy over a 30 s run is 90.1%. The scene includes a crossing pedestrian actor, which is labeled as an obstacle when it approaches the robot path; the human handler follows the robot at a fixed offset. Fig. 2 shows the rectified forward view.

Table 2 reports the pose RMSE over the run. The standard particle filter uses the base covariance, while SAPF uses $\mathbf{Q}(C_t)$. SAPF reduces the Z-axis RMSE from 0.150 m (SPF) to 0.024 m, and the overall position RMSE from 0.152 m to 0.030 m. On the stair segment alone, the Z-axis RMSE is 0.474 m for WO, 0.234 m for SPF, and 0.022 m for SAPF. Fig. 3 and Fig. 4 show the X–Z trajectory and the terrain labels used by the filter.

Table 1: AHPFF clustering ablation on the corrected NYU Depth V2 pilot subset (15 samples per class, degraded setting). ACC and NMI are percentages.

Method	ACC (%)	NMI (%)
Vision only	51.1	10.0
Depth/LiDAR only	55.6	23.0
Fixed concatenation	66.7	28.1
AHPFF	**71.1**	**36.7**

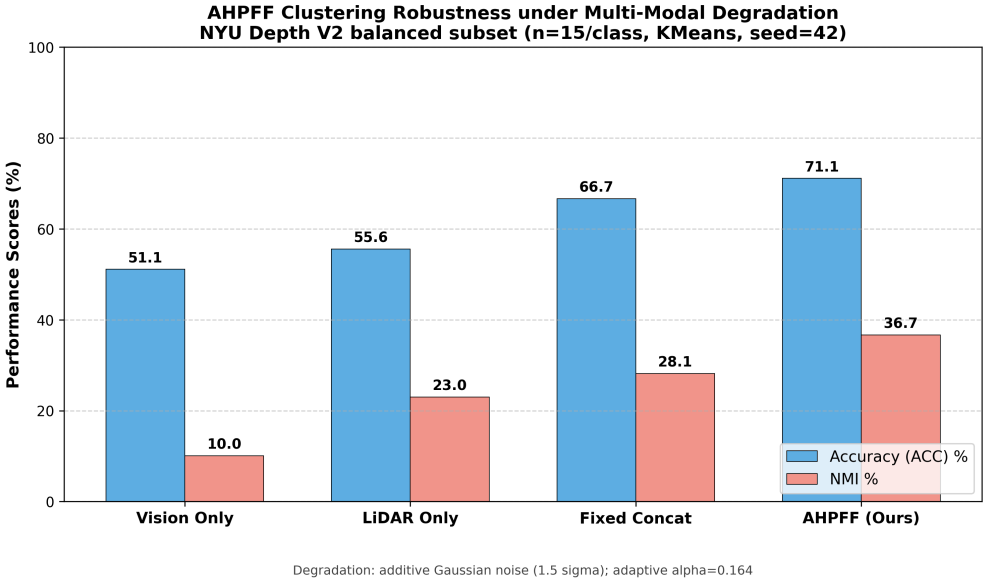


Figure 1: AHPFF clustering under multi-modal degradation on the corrected NYU Depth V2 pilot subset. The subset is balanced to 15 samples per class because annotated stair frames are scarce.

Table 2: Go1 closed-loop pose RMSE in the Gazebo street scene (30 s, 50 Hz, 500 particles). Values are in meters.

Method	X	Y	Z	Overall
Wheel odometry (WO)	0.087	0.153	0.325	0.370
Standard PF (SPF)	0.015	0.016	0.150	0.152
SAPF (Ours)	**0.014**	**0.013**	**0.024**	**0.030**

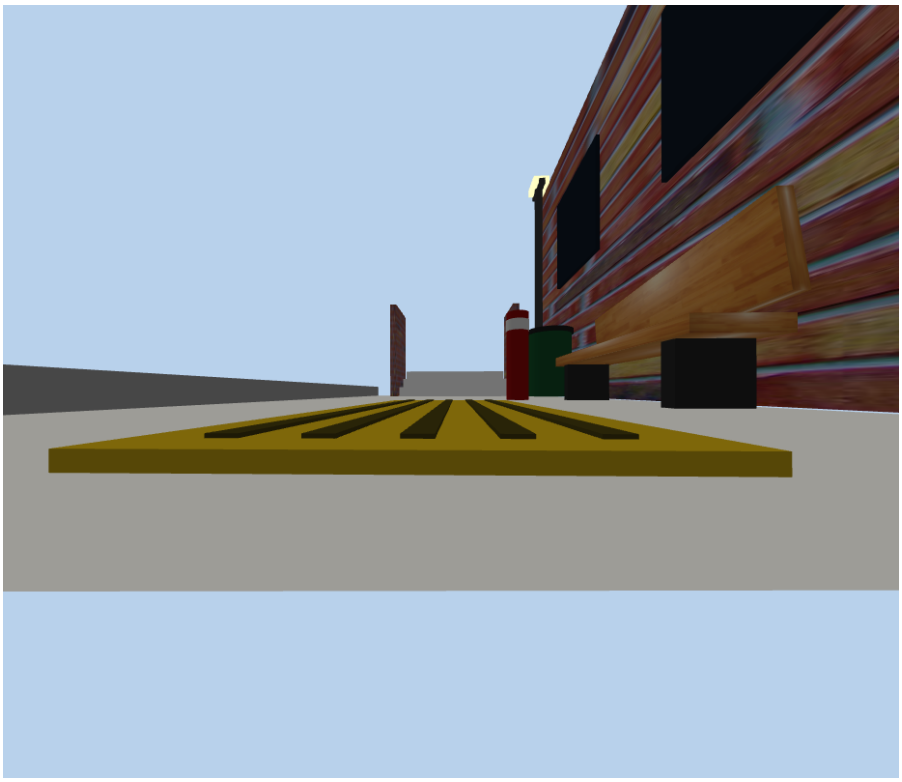


Figure 2: Rectified forward RGB-D view of the Go1 street scene. The scene contains tactile paving, a bollard, a walking pedestrian actor, street furniture, and a staircase.

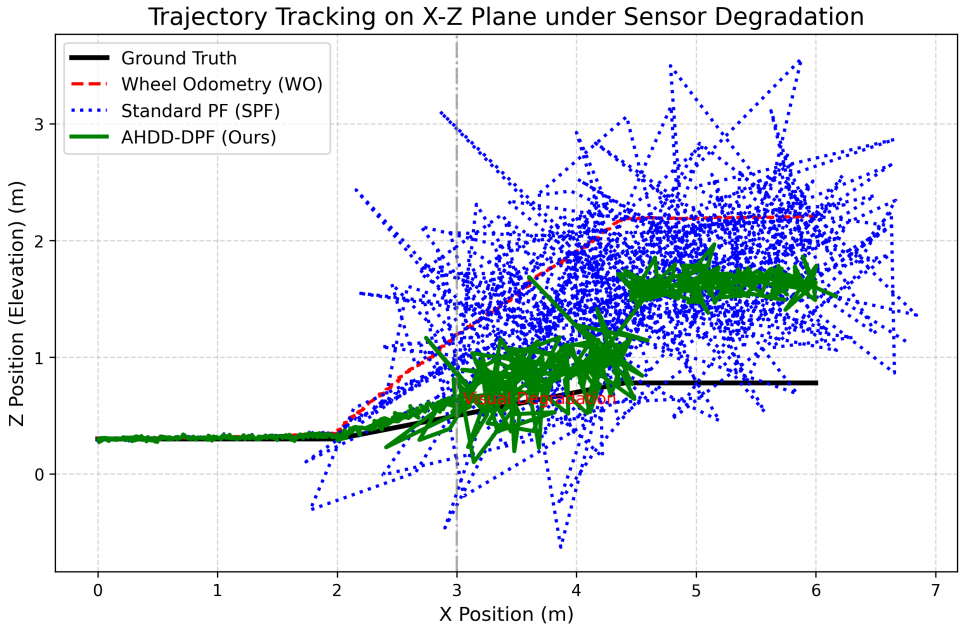


Figure 3: X–Z trajectory in the Go1 closed-loop study. WO drifts on the stair segment; SPF lags; SAPF tracks the elevation changes.

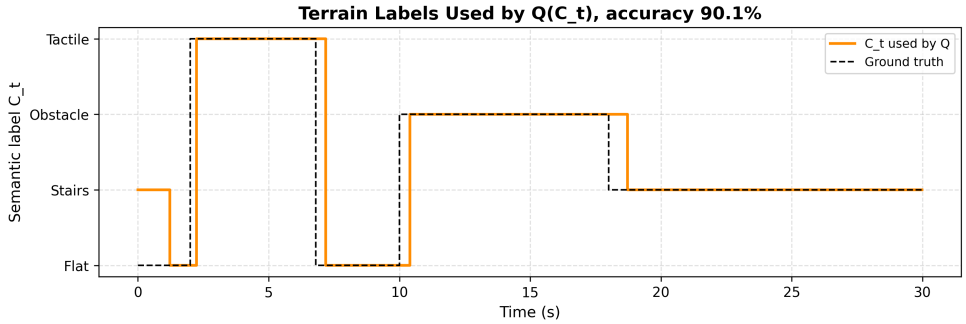


Figure 4: Terrain label C_t used by SAPF versus the scripted ground-truth label. The online semantic accuracy is 90.1% over the run.

4.3. Sensor Degradation Study

We inject overexposure, horizontal motion blur, and 50% random depth dropout in the perception node from 6 to 20 s of the run. The Gazebo scene itself is unchanged; the degradation affects the RGB-D stream and the AHPFF features. Fig. 5 shows a degraded preview. Table 3 reports a separate baseline/degraded pair; its baseline semantic accuracy (90.8%) differs slightly from the live run in Section 4.2 (90.1%) because of run-to-run variation. The table compares the baseline and degraded runs. Semantic accuracy falls from 90.8% to 71.5%, and SPF's Z-axis RMSE worsens from 0.154 m to 0.190 m. SAPF keeps the Z-axis RMSE at 0.022 m in both runs, and the overall position RMSE remains about 0.029 m. This is the main empirical support for the semantic-adaptive covariance: when the label stream becomes unreliable, the expanded vertical/pitch process noise still allows the filter to track the elevation changes.

Table 3: Sensor degradation study on the Go1 closed-loop system. The degraded run applies overexposure, motion blur, and 50% depth dropout from 6 to 20 s.

Run	Semantic ACC	WO Z	SPF Z	SAPF Z	SAPF Overall
Baseline	90.8%	0.325	0.154	0.022	0.0285
Degraded	71.5%	0.408	0.190	0.022	0.0286

4.4. Discussion and Limitations

The results support the AHPFF–SAPF design, but several limitations must be stated. First, the NYU study is a 15-sample-per-class pilot subset because annotated stair frames are scarce; it is not a large-scale benchmark. Second, the online C_t is produced by a supervised ExtraTrees readout on the AHPFF descriptor bank, not by assigning unsupervised cluster indices; silhouette weighting is used only in the offline ablation. Third, the $\mathbf{Q}(C_t)$ boosts are configured/calibrated values rather than learned parameters. Fourth, the Gazebo observation is a simulated noisy absolute pose and the Go1 motion is kinematic; the study does not constitute a physics-based locomotion or a real-world deployment. Fifth, the classifier is trained and tested in the same Gazebo scene, so the reported 98.29% cross-validation accuracy is an in-scene result, not evidence of cross-domain generalization. These limitations define the scope of the current claims and the next steps toward a physical Go1 controller and a real-world validation.

5. Conclusion

We presented AHPFF and SAPF for robust terrain perception and 6-DoF state estimation in robotic guide dogs. AHPFF uses hand-crafted visual and depth descriptors, silhouette-weighted fusion in the clustering ablation, and a lightweight supervised semantic readout. SAPF tracks the full 6-DoF state, adapts its process covariance to the terrain label, and uses systematic resampling. On a corrected NYU pilot subset, AHPFF improves degraded clustering over fixed concatenation and single views. In a Go1 Gazebo closed loop with a forward RGB-D camera, a dynamic

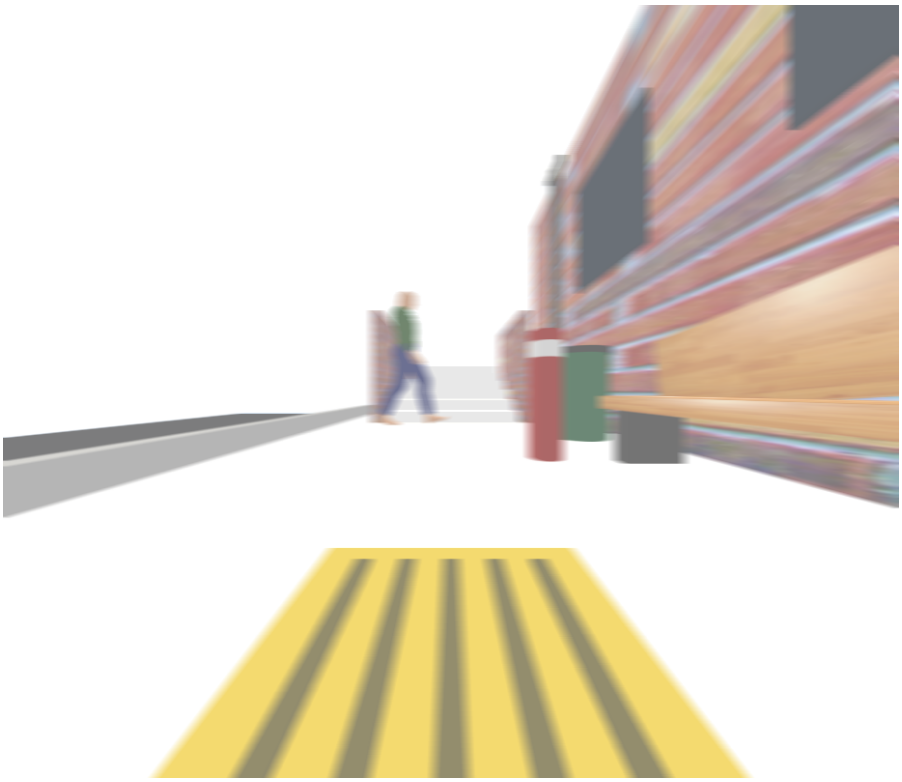


Figure 5: Degraded RGB-D preview used in the sensor-degradation study. The image shows overexposure and horizontal motion blur; the walking pedestrian is still visible.

pedestrian, and a scripted handler, the system reaches 90.1% semantic accuracy and reduces the Z-axis RMSE from 0.150 m (SPF) to 0.024 m; under severe sensor degradation, SAPF keeps the Z-axis RMSE at 0.022 m while SPF degrades. The current study is limited to a pilot NYU subset, kinematic Gazebo locomotion, simulated observations, and in-scene classification. Future work will integrate a physics-based locomotion controller, evaluate on a real quadrupedal platform, and study cross-domain terrain generalization.

Acknowledgments

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